

Effects of neutral detergent fiber content and its source on growth performance, rumen characteristics, digestibility, and methane concentrations in Hanwoo heifers

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ABSTRACT

Objective: This study investigated the effects of neutral detergent fiber (NDF) content and NDF source (forage- or concentrate-derived) on growth performance, rumen characteristics, nutrient digestibility, and methane (CH₄) concentrations in growing Hanwoo heifers.

Methods: Thirty heifers (296 ± 25.7 kg body weight [BW]; 10 months old) were blocked by BW and assigned to three treatments: high forage-to-concentrate ratio (42.5:57.5) with high-energy concentrate (HFHC), low forage-to-concentrate ratio (30:70) with low-energy concentrate (LFLC), and low forage-to-concentrate ratio (30:70) with high-energy concentrate (LFHC). NDF-level effects were assessed by contrasting HFHC and LFLC with LFHC, whereas the NDF-source contrast compared HFHC with LFLC at similar total NDF levels. Dry matter intake (DMI) was recorded daily and BW every four weeks. Rumen fluid and fecal samples were collected, and CH₄ concentrations measured, after 12 weeks.

Results: Average daily gain (ADG) and feed conversion ratio were unaffected by NDF level or source ($P > 0.05$), whereas DMI tended to increase with lower NDF ($P = 0.099$). Lower NDF reduced NDF digestibility and increased total volatile fatty acid concentration ($P < 0.05$), decreased acetate proportion and acetate-to-propionate ratio, and increased propionate and butyrate proportions ($P < 0.001$). NDF source did not affect nutrient digestibility ($P > 0.05$). In the NDF-source contrast, LFLC had lower acetate and propionate and higher butyrate proportions than HFHC ($P < 0.05$). Lower NDF also decreased CH₄ concentration per DMI

and ADG ($P < 0.05$). LFLC had lower CH₄ concentration per ADG than HFHC, whereas CH₄ concentration per DMI did not differ.

Conclusion: Dietary NDF level was associated with broader changes in ruminal fermentation, NDF digestibility, and CH₄ concentration per DMI and ADG, whereas NDF source had more limited effects, primarily on VFA proportions and CH₄ concentration per ADG. Neither NDF level nor source significantly affected growth performance in growing Hanwoo heifers.

Keywords: methane, forage-derived neutral detergent fiber, concentrate-derived neutral detergent fiber, laser methane detector, Hanwoo

INTRODUCTION

Alongside global efforts to reduce greenhouse gases, reducing methane (CH₄) emissions from ruminants, which account for 16% of global CH₄ emissions [1], and represent a loss of 2–12% of gross energy intake in ruminants [2], remains an important area of research in ruminant nutrition. Various strategies have been developed for reducing CH₄ emissions in ruminants through animal and feed management, dietary strategies, and manipulation of the rumen, with numerous studies conducted on these approaches [3,4].

Dietary strategies are considered practical approaches that can reduce CH₄ emissions from ruminants while minimizing negative impacts on productivity [5]. Various dietary strategies, including the supplementation of oils or fats, adjusting feed ratios, and additives, have been evaluated for their effect on reducing CH₄ emissions [6,7]. Particularly, reducing dietary neutral detergent fiber (NDF) levels or improving its digestibility has been considered an effective nutritional strategy to mitigate CH₄ emissions [8]. Lowering dietary NDF levels by increasing the proportion of concentrate mix in the diet can result in lower CH₄ emissions compared with high-forage diets characterized by high dietary NDF levels [2,9]. This reduction may be partly associated with the fermentation characteristics of starch contained in grains. Unlike structural carbohydrates in forages, rapid starch degradation can shift ruminal fermentation pathways toward propionate and butyrate production, which may reduce the availability of metabolic H₂ for methanogenesis [10–12]. Conversely, diets high in fibrous forages are generally associated with greater acetate production and H₂ generation, which can favor CH₄ formation [13]. Beyond fermentation pathways, the physical kinetics of digestion may also influence CH₄ production. High-concentrate intake can increase the passage rate of digesta and may shorten ruminal retention time, thereby limiting the extent of microbial fermentation and potentially contributing to lower CH₄ emissions [3,14]. Additionally, rapid fermentation of grains can

decrease ruminal pH, which may suppress methanogenic activity and alter protozoal populations, potentially contributing to reduced CH₄ formation [15].

Various studies have investigated the effects of varying dietary NDF levels through the adjustment of forage-to-concentrate (FC) ratios on CH₄ emissions. Aguerre et al. [16] reported that reducing the forage proportion from 68% to 47% in lactating Holstein cows, which lowered dietary NDF levels from 39% to 31% dry matter (DM), resulted in an approximately 17% decrease in CH₄ emissions ($P < 0.05$). Similarly, Olijhoek et al. [17] observed that reducing the forage ratio from 70% to 40% in Holstein and Jersey dairy cows, thereby decreasing NDF levels by approximately 6% (from 34% to 27% DM), led to a reduction in average CH₄ emissions of about 10% ($P < 0.05$). These results were also observed in beef cattle. Lovett et al. [18] found that increasing the proportion of concentrate mix from 35% to 90% in finishing heifers, which reduced NDF levels by approximately 14% DM, increased average daily gain (ADG) by 0.4 kg and decreased CH₄ emissions by approximately 18% ($P < 0.01$). Kang et al. [19] also reported that increasing the concentrate mix proportion by 10% (from 65% to 75%) in growing Hanwoo steers, which reduced NDF levels by 11% DM, decreased CH₄ emissions via respiration and eructation by approximately 33% and 36%, respectively ($P < 0.05$). Furthermore, according to a meta-analysis by Arndt et al. [3], reducing the FC ratio (i.e., decrease NDF level in diets) led to a 9% decrease in CH₄ emissions per milk production and weight gain. These results are attributed to the fact that a lower FC ratio increases feed intake, milk production, and weight gain without increasing CH₄ emissions.

Although the effects of reducing dietary NDF through adjustment of the FC ratio on CH₄ emissions are well established, most previous studies have focused primarily on changes in total NDF content, largely through manipulation of the FC ratio. However, changes in forage inclusion are often accompanied by simultaneous changes in both total NDF content and the

relative contribution of forage-derived NDF. In addition, as the inclusion of fibrous ingredients such as agricultural by-products in concentrate mixes has increased, a substantial proportion of total dietary NDF may originate from the concentrate fraction rather than from forage [20]. Forage-derived NDF (fNDF) and concentrate-derived NDF (cNDF) may differ markedly in physical effectiveness, fermentability, and their effects on H₂ metabolism, a precursor process for methanogenesis [14,21]. Unlike chemically determined NDF concentration, physical effectiveness reflects the ability of fiber to stimulate chewing and contribute to ruminal mat formation and is influenced by physical characteristics such as particle size [21]. Considering both dietary NDF level and the predominant source of NDF may provide a more comprehensive understanding of their relationships with ruminal fermentation, productivity, and CH₄ emissions and may contribute to the development of effective dietary strategies for CH₄ mitigation. Despite this relevance, relatively few studies have considered dietary NDF level and predominant NDF source within the same experimental framework, and, to our knowledge, no such studies have evaluated these two NDF characteristics together in growing Hanwoo heifers. Therefore, the objective of this study was to investigate the effects of NDF content and its source on growth performance, rumen characteristics, nutrient digestibility, and enteric CH₄ concentrations in growing Hanwoo heifers.

MATERIALS AND METHODS

This study was conducted at the Center for Animal Science Research, Chungnam National University, Republic of Korea. The use of animals and protocols for this experiment were reviewed and pre-approved by the Chungnam National University Animal Research Ethics Committee (202109A-CNU-108).

Animals, housing, and diets

Thirty 10-month-old Hanwoo heifers (296 ± 25.7 kg body weight [BW]) were used in this experiment. The heifers were stratified into five blocks based on initial BW, with each block consisting of six heifers housed in three pens. Two heifers of similar BW were housed together in each pen ($5 \text{ m} \times 5 \text{ m}$), and the three pens within each block were randomly assigned to one of the three dietary treatments according to a randomized complete block design [22], resulting in five pens per treatment. Each pen was equipped with one TMR feed bunk that allowed individual feed intake to be recorded automatically by identifying each animal using a radio-frequency identification tag (Dawoon Co., Incheon, Korea). The experiment lasted for three months, after a seven-day adaptation period.

The diet was prepared as a total mixed ration (TMR) based on oat hay, perennial ryegrass, corn, corn gluten feed, palm kernel meal, and other ingredients. The TMR was adjusted to vary the FC ratio, with the energy levels being regulated through low energy concentrate mix (LC) and high energy concentrate mix (HC). There were three dietary treatments: 1) high FC ratio with HC (HFHC, 42.5% forage, and 57.5% HC), 2) low FC ratio with LC (LFLC; 30% forage and 70% LC), and 3) low FC ratio with HC (LFHC; 30% forage and 70% HC). The diets were formulated to meet nutrient requirements for target ADG of 650 g/d for both the HFHC and LFLC treatments and 800 g/d for the LFHC treatment, according to the Korean feeding standards for Hanwoo heifers [20]. The ingredient combinations were selected to create contrasts in total NDF level and predominant NDF source. HFHC combined a higher forage proportion with a high-energy concentrate, whereas LFLC contained more fibrous concentrate ingredients, particularly palm kernel meal, to achieve a similar total NDF concentration. LFHC combined the lower forage proportion with a high-energy concentrate to achieve a lower NDF concentration. The higher target ADG for LFHC reflected its accompanying energy density

rather than an independently imposed growth treatment. Consequently, the NDF-level contrast also involved differences in energy supply, and the NDF-source contrast involved differences in ingredient and nutrient composition; neither contrast isolated NDF as the sole dietary factor.[23] The formulation and chemical composition of the experimental diets are presented in Tables 1 and 2, respectively. Heifers were fed twice daily at 0800 and 1800 h. The TMR and drinking water were freely accessible to the animals throughout the experiment. During the experiment, one heifer (LFLC treatment) was excluded from the study due to health issues and was therefore omitted from all sampling and data analysis.

Measurement and sample collection

The individual daily intake was recorded individually through the TMR feed bunk (Dawoon, Co., Incheon, Korea). The TMR feed bunk recognizes the heifer's neck radio-frequency identification tag and automatically records the individual intake by measuring the weight difference before entry and after exit of the heifer, using two installed load cells. Every four weeks, all daily feed intake records for each heifer were collected and processed separately. The mean and standard deviation were calculated from the daily intake records of each heifer during each four-week period. Intakes that were more or less than three times the standard deviation from the mean intake were treated as outliers and removed. The intakes of the day when management activities took place that could have affected feed intakes, such as bedding replacement, BW measurement, and sampling periods, were also removed. The BW was measured every four weeks before morning feeding, and the feed samples for chemical composition analysis were sampled once every four weeks.

After 12 weeks, 15 heifers (five per treatment) were randomly selected within each dietary treatment immediately before the digestibility sampling period. Selection was performed at the

individual-animal level and was not stratified by pen, initial BW, or previous feed intake. Only heifers remaining in the experiment at the time of selection were eligible for sampling. Fecal samples were collected by spot sampling eight times over four consecutive days at nine-h intervals (d 1, 17:00; d 2, 02:00, 11:00, and 20:00; d 3, 05:00, 14:00, and 23:00; and d 4, 08:00). The collected fecal samples were dried at 65 °C for 72 h, and the dried samples from the eight sampling time points were pooled on an equal-weight basis within each heifer for subsequent analyses.

Rumen fluid of all heifers was collected three times (−1, +3, and +6 h after morning feeding) on three consecutive days after 12 weeks using an oral stomach tube, as described by Lee et al. [24] and Oh et al. [25]. Briefly, the initially collected rumen fluid was discarded (approximately 300 mL), and 400 mL of rumen fluid was subsequently collected in a glass flask. After collection, the pH of the rumen fluid was immediately measured, and 10 mL each was subsampled for ammonia (NH₃-N) and volatile fatty acid (VFA) analysis and stored at -20 °C until analysis.

Methane concentrations of all heifers were measured four times daily (−2, −1, +1, and +2 h after morning feeding) using a laser methane detector (LMD; Laser Methane Mini, Tokyo Gas Engineering Solutions Co. Ltd, Tokyo, Japan) for five consecutive days after 12 weeks, as described by Kang et al. [19] and Oh et al. [25]. Briefly, with the LMD installed stably on a tripod, the visible laser was aimed at the heifer's nose from a distance of 1 m, and CH₄ concentrations were measured every 0.5 s for 6 min. The measurements were duplicated, and peak values were identified separately for each measurement. CH₄ concentrations were calculated independently from each measurement, and the mean of the two measurements was used as the individual animal value for subsequent statistical analysis. The LMD measures path-integrated CH₄ concentration in ppm·m [16]. At the fixed measurement distance of 1 m,

division by the path length gives a path-averaged concentration in ppm with the same numerical value. These concentration-based measurements were not converted into methane mass or volume per day.

Chemical analyses and data processing

Chemical analyses were performed as described by Jeon et al. [26]. The feed and fecal samples were dried at 60 °C for 96 h and ground through a cyclone mill (Foss, Hillerød, Denmark) fitted with a 1 mm screen. The nutrient composition of the feed samples was analyzed at Cumberland Valley Analytical Services Inc (Hagerstown, MD, USA). The DM content (#934.15), crude protein (#990.03), ether extract (#920.39), acid detergent fiber (#973.18), and ash content (#942.05) were measured. Crude protein was estimated by multiplying the nitrogen content by 6.25, with nitrogen quantified using the Dumas method, on a Leco FP-528 Nitrogen Combustion Analyzer (Leco Inc., Saint Joseph, MI, USA). The acid detergent lignin (ADL) was measured, and NDF contents were analyzed using a heat-stable amylase, including residual ash. Additionally, soluble protein, neutral-detergent-insoluble crude protein (NDICP), and acid-detergent-insoluble crude protein (ADICP) were determined. The contents of macro- and micro-minerals were determined.

The total digestible nutrients (TDN) content was estimated according to the dairy NRC [27]. Based on TDN, metabolizable energy (ME), net energy for maintenance (NE_m), and net energy for growth (NE_g) were calculated using the equations described in the beef NRC [28]. Dietary carbohydrates and protein fractions were estimated according to Cornell Net Carbohydrate and Protein System [29] with the following modifications. For carbohydrate fractions, CA represents sugars and organic acids assumed to be equal to ESC, CB1 represents starch, CB2 represents soluble fiber and calculated as non-fiber carbohydrates (NFC) – CA – CB1, CB3

represents available NDF and estimated by $\text{NDF} - \text{NDICP}$ minus 2.4 times ADL, and CC represents unavailable carbohydrate and estimated by 2.4 times ADL. For protein fractions, PA+B1 represents soluble protein and equals to SOLP, PB2 represents intermediate degradable CP and estimated by $100 - \text{NDICP} - \text{SOLP}$, PB3 represents slowly degradable fiber-bound CP and estimated by $\text{NDICP} - \text{ADICP}$, and PC represents unavailable CP and equals to ADICP. All the carbohydrate and protein fractions were expressed as g/kg of total carbohydrate or CP, respectively

The $\text{NH}_3\text{-N}$ concentration of the rumen fluid was determined as follows. Following centrifugation of the rumen fluid at $21,000 \times g$ for 15 min, 20 μL of the supernatant was mixed with 1 mL of phenol color reagent and 1 mL of alkali hypochlorite reagent. The mixture was then incubated in a water bath for 15 min at 37°C . After being mixed with 8 mL of distilled water, the optical density of the mixture was measured at 630 nm using a spectrophotometer (UV-1800, Shimadzu, Kyoto, Japan).

In order to determine the VFA concentration of rumen fluid, rumen fluid supernatant (1 mL) was mixed with 0.2 mL of metaphosphoric acid (250 g/L) and kept at 4°C for 30 min. Following centrifugation of the mixture at $21,000 \times g$ for 10 min at 20°C , the supernatant was injected into a gas chromatograph (HP 6890, Hewlett-Packard Co., Palo Alto, CA, USA) equipped with a flame ionization detector and capillary column (Nukol Fused silica capillary column $30\text{ m} \times 0.25\text{ mm} \times 0.2\text{ }\mu\text{m}$, Supelco Inc., Bellefonte, PA, USA). The temperature of the oven, injector, and detector was 90°C to 180°C , 185°C , and 210°C , respectively. Nitrogen was used as the carrier gas at a flow rate of 40 mL/min.

Nutrient digestibility was determined using acid-insoluble ash (AIA) as an internal marker, as described in Van Soest et al. [30]. The AIA concentrations in feed and feces were analyzed

using the method described by Van Keulen and Young [31]. Nutrient digestibility was calculated as follows:

$$\text{Nutrient digestibility} = 100 - 100 \times \frac{\text{AIA in feed} \times \text{nutrient in feces}}{\text{AIA in feces} \times \text{nutrient in feed}}$$

where AIA in feed and AIA in feces represent the AIA contents of the feed and feces, respectively, and nutrient in feed and nutrient in feces represent the contents of the corresponding nutrient in the feed and feces, respectively. All values were expressed on a DM basis.

Methane concentration data were separated into respiration and eructation by detecting CH₄ concentration peaks using the automatic multi-scale peak detection package in R software [32] and fitting a double normal distribution using the mixdist package in R software. The mean of a normal distribution represents the mean CH₄ concentration per day as the mean of four time-of-day values, assuming representative CH₄ concentrations of exhaled gas from the pathway during that time period [19].

Statistical analysis

All data were analyzed using the PROC MIXED procedure of SAS (SAS Institute Inc., Cary, NC, USA). The blocks (i.e., initial BW) were treated as random effects.

$$y_{ijk} = \mu + \tau_i + b_j + e_{ijk}$$

where y_{ijk} is the mean observation of the k th animal receiving the i th treatment within the j th block, μ is the overall mean, τ_i is the fixed effect of the i th treatment ($i = 1-3$), b_j is the random effect of the j th block ($j = 1-5$), and e_{ijk} is the residual error associated with the k th animal receiving the i th treatment within the j th block.

Orthogonal contrasts were performed to test the differences between NDF levels (LFHC vs. the average of HFHC and LFLC) and between NDF sources (HFHC vs. LFLC). For HFHC,

LFLC, and LFHC, respectively, the NDF-level contrast coefficients were -0.5 , -0.5 , and 1.0 , and the NDF-source contrast coefficients were 1.0 , -1.0 , and 0 . These coefficients were specified a priori and were not reweighted after one LFLC heifer was excluded. The mixed model used all available observations to estimate treatment effects in the unbalanced dataset. Individual means were also compared using Tukey's test. To test the effects of the different treatments on the rumen parameters, the data were analyzed as repeated measures to account for the correlation between repeated measurements of each animal. For this analysis, no structure was assumed for the variance-covariance matrix. Significance was declared at $P < 0.05$, and a trend was discussed at $0.05 \leq P < 0.1$.

RESULTS

Neither NDF level nor NDF source had a significant effect on ADG and feed conversion ratio (FCR, DMI [g]/ADG [g]; Table 3; $P > 0.05$). However, dry matter intake (DMI) tended to increase as the dietary NDF level decreased ($P = 0.099$; DMI means of 7.17 , 7.46 , and 8.02 kg/day for HFHC, LFLC, and LFHC, respectively), while the NDF source did not significantly influence DMI ($P > 0.05$).

Neither NDF level nor NDF source affected rumen fluid pH (Table 4; $P > 0.05$). The overall treatment effect on $\text{NH}_3\text{-N}$ concentration was significant ($P = 0.047$), with higher values in LFLC and LFHC than in HFHC (11.1 and 11.7 vs. 7.6 mg/dL, respectively; Table 4). In the planned contrasts, $\text{NH}_3\text{-N}$ concentration was not affected by NDF level ($P > 0.05$), but tended to be affected by NDF source ($P = 0.052$). Decreasing NDF level significantly affected total VFA concentration, with LFHC exhibiting the highest VFA concentration (Table 4; $P < 0.05$). Reducing the dietary NDF level significantly altered VFA proportions (Table 4; $P < 0.001$). A lower NDF level led to a significant decrease in the proportion of acetate ($P < 0.001$), resulting

in the lowest acetate proportion in the LFHC group (616.8 mmol/mol). Conversely, decreasing the NDF level significantly increased the proportion of propionate ($P < 0.001$), with the LFHC group showing the highest proportion (213.9 mmol/mol). As a result of these shifts, the acetate-to-propionate (A:P) ratio significantly decreased as the NDF level declined ($P < 0.001$), with the LFHC group exhibiting the lowest ratio (2.9). Additionally, a lower NDF level significantly increased the proportion of butyrate ($P < 0.001$). The NDF source also significantly affected VFA composition (Table 4; $P < 0.05$). In the NDF-source contrast, LFLC had significantly lower molar proportions of both acetate ($P = 0.018$) and propionate ($P = 0.007$). Consequently, the LFLC group exhibited lower proportions of acetate (649.7 vs. 669.3 mmol/mol) and propionate (187.4 vs. 203.1 mmol/mol) compared to the HFHC group. In contrast, the butyrate proportion was significantly higher in LFLC than in HFHC ($P < 0.001$), as evidenced by the higher butyrate proportion in the LFLC group (116.6 mmol/mol) compared to the HFHC group (90.7 mmol/mol). However, unlike the NDF level, the shift in NDF source did not result in a significant difference in the A:P ratio ($P = 0.128$). Furthermore, the proportions of isobutyrate and isovalerate were significantly higher in the LFLC group than in the HFHC group ($P < 0.01$), while valerate tended to be higher in the LFLC group ($P = 0.065$).

Decreasing dietary NDF level reduced nutrient digestibility (Table 5). Digestibility of NDF decreased as dietary NDF level decreased ($P = 0.012$), while ADF digestibility tended to decrease ($P = 0.053$). CP digestibility was also lower in the low-NDF group ($P = 0.013$), and EE digestibility tended to decrease with decreasing NDF level ($P = 0.065$). Regarding NDF source, DM and ADF digestibility tended to be lower in the LFLC group than in the HFHC group ($P = 0.072$ and $P = 0.095$, respectively), whereas the digestibility of the other nutrients was not affected by NDF source ($P > 0.05$).

Methane concentration in the exhaled gas from respiration and eructation decreased as NDF levels decreased, with the LFHC treatment recording the lowest values of 16.4 ppm and 82.7 ppm, respectively (Table 6; $P < 0.05$). Additionally, in the NDF-source contrast, CH₄ concentrations were lower in LFLC than in HFHC; the cNDF-based LFLC treatment showed significant reductions of approximately 19% in both respiration and eructation compared to the fNDF-based HFHC treatment ($P < 0.05$). Methane concentration per DMI (ppm/kg) significantly decreased as NDF levels decreased (Table 6; $P < 0.05$), with the LFHC treatment showing the lowest values for respiration (2.1 ppm/kg of DMI) and eructation (10.5 ppm/kg of DMI). However, no significant difference was observed between NDF sources regarding CH₄ concentration per DMI (ppm/kg) from either respiration or eructation ($P > 0.05$). In contrast, CH₄ concentration per ADG (ppm/kg) was significantly influenced by both NDF level and source. Methane concentration per ADG (ppm/kg) decreased as NDF levels decreased, with the LFHC treatment recording the lowest values of 21.1 and 107.7 ppm/kg of ADG for respiration and eructation, respectively ($P < 0.05$). Regarding NDF source, the cNDF-based LFLC treatment exhibited significantly lower CH₄ concentration per ADG (ppm/kg) compared to the fNDF-based HFHC treatment, showing reductions of approximately 28% for respiration and 30% for eructation ($P < 0.05$).

DISCUSSION

Reducing CH₄ emissions from ruminants is a primary focus of global greenhouse gas mitigation efforts. Nutritional strategies, particularly the modulation of dietary NDF levels, have been widely recognized as effective methods to reduce CH₄ emissions. However, most previous studies have focused primarily on total NDF levels or FC ratios, often overlooking potential differences arising from the source of the fiber. Since fNDF and cNDF possess distinct

physicochemical and fermentative characteristics, they may influence rumen fermentation and methanogenesis differently. Therefore, this study aimed to investigate the effects of not only NDF levels but also NDF sources on growth performance, rumen characteristics, nutrient digestibility, and CH₄ concentrations in growing Hanwoo heifers.

It is generally accepted that DMI in ruminants is negatively correlated with dietary NDF concentration due to the physical filling effect [33,34]. In the present study, although the effect was not significant, DMI tended to increase as dietary NDF level decreased ($P = 0.099$), with no difference between NDF sources ($P > 0.05$). The relatively small increase in DMI observed in the low-NDF group may reflect a greater contribution of metabolic regulation of feed intake relative to physical fill [35]. Although the LFHC group was fed the diet with the lowest NDF level, which could be expected to reduce physical fill in the rumen, its higher energy density may have promoted metabolic satiety and thereby limited further increases in feed consumption [33]. Previous research has also reported that DMI was not affected by increased NDF level in animals consuming high-energy diets, despite variations in dietary starch and NDF levels [16,19].

Generally, reducing dietary NDF levels implies an increase in the proportion of readily fermentable carbohydrates, which is expected to stimulate rumen fermentation and improve nutrient digestibility, potentially leading to enhanced growth performance [34]. Serment et al. [36] reported that organic matter (OM) digestibility improved by approximately 6% ($P < 0.05$) in goats when dietary NDF levels were decreased by adjusting the concentrate mix proportion of the TMR from 30% to 70% DM. Similarly, Chen et al. [34] observed an improvement in OM digestibility (59% vs. 67% DM; $P < 0.05$) and an increase in ADG (0.75 vs. 1.1 kg/day; $P < 0.05$) in Angus cows when dietary NDF levels were decreased by raising the dietary concentrate mix ratio by 30%. However, in this study, despite differences in dietary NDF levels

as well as dietary energy levels, ADG did not differ among treatments ($P > 0.05$). Moreover, decreasing NDF levels significantly reduced the digestibility of CP and NDF (Table 5; $P < 0.05$). The lower nutrient digestibility may have contributed to the absence of a detectable growth response despite the higher dietary energy density; however, this association does not establish a causal relationship. Although ruminal retention time was not directly measured in the present study, one possible explanation for the lower fiber digestibility observed at the lower NDF level is a reduction in ruminal retention time. Dietary NDF plays a critical role in retarding the flow of digesta and forming a physical matrix within the rumen, providing sufficient time for microbes to attach to and degrade feed particles [37]; however, when dietary NDF levels are reduced, as in the LFHC group, ingested feed may pass through the rumen more quickly, potentially reducing the time available for microbial fermentation. Therefore, a faster passage rate could have reduced the time available for microbial degradation, particularly for fiber, which generally requires a longer period for digestion [38]. Furthermore, the high NFC level corresponding to low NDF levels may have induced a negative associative effect on the ruminal microbial ecosystem. When the proportion of rapidly fermentable carbohydrates is high, starch-degrading bacteria may dominate in substrate competition against fiber-degrading bacteria or cause acidification of the local micro-environment, thereby inhibiting the activity of fibrolytic enzymes. Furthermore, one possible explanation for the reduction in CP digestibility observed at lower NDF levels is increased fecal nitrogen excretion resulting from shifts in the site of fermentation. Low dietary NDF levels may accelerate the passage rate, reducing the extent of carbohydrate fermentation in the rumen and allowing more fermentable carbohydrates to reach the large intestine [39]. These shifts may promote microbial growth in the hindgut. While microbial protein synthesized in the rumen is largely digested and absorbed in the small intestine with only a small portion excreted, microbial protein synthesized in the

hindgut is completely excreted in the feces [40]. Since apparent digestibility calculations do not distinguish between undigested dietary nitrogen and metabolic fecal nitrogen (including microbial debris), a possible increase in fecal microbial nitrogen could lower apparent CP digestibility without demonstrating a reduction in dietary protein digestion. Passage rate, site of digestion, and fecal microbial nitrogen were not measured in this study.

In this study, reducing the NDF level increased the total VFA concentration, whereas it decreased the proportion of acetate and increased those of propionate and butyrate (Table 4; $P < 0.05$). These changes are attributed to the high NFC level in the low-NDF diet. Compared to structural carbohydrates, NFC are degraded more rapidly and extensively by rumen microbes, thereby increasing the fermentation rate and the total production of organic acids [41]. Furthermore, high dietary NFC levels promote starch degradation in the rumen and shift fermentation pathways toward propionate production, consequently decreasing acetate and increasing propionate proportion [14]. These shifts are consistent with changes in the availability of fermentation substrates, as reduced fiber availability and increased fermentable starch supply generally favor lower acetate and greater propionate production [42]. Moreover, the molar proportion of butyrate increased as NDF levels declined, which is a typical response to diets rich in NFC. Rapid fermentation of NFC can increase lactate production, which can subsequently be converted to butyrate through established ruminal fermentation pathways [42]. Previous studies [34,36,43,44] have also reported results consistent with the findings of the present study.

In addition to the NDF level, the source of NDF also significantly influenced the ruminal VFA fermentation profile. Specifically, the group fed cNDF (LFLC) exhibited significantly lower proportions of acetate and propionate compared to the group fed fNDF (HFHC; Table 4; $P < 0.05$). Conversely, the proportion of butyrate was significantly higher in the LFLC group

($P < 0.001$). Interestingly, despite the simultaneous reduction in acetate and propionate in the LFLC group, there was no significant difference in the A:P ratio between the two groups ($P > 0.05$). These results indicate that the LFLC diet was associated with a shift in the overall fermentation pattern toward alternative VFAs without disrupting the balance between acetate and propionate. The higher acetate proportion observed in the HFHC group may be associated with its high forage (fNDF) level, as structural carbohydrates in forage are primarily fermented into acetate by cellulolytic bacteria [45]. Dietary starch can favor propionate formation [11], but the analyzed starch concentration was not higher in HFHC than in LFLC (136 vs. 148 g/kg DM; Table 2). Therefore, starch concentration alone does not explain the higher propionate proportion in HFHC. Differences in ingredient composition and carbohydrate availability may also have contributed to the higher butyrate proportion in LFLC.[14] These results appear to be attributable to the fact that the cNDF in the LFLC diet was primarily composed of palm kernel meal (331 g/kg DM). Palm kernel meal is a representative fibrous by-product containing high levels of hemicellulose, particularly mannans [46]. These hemicellulosic substrates serve as a preferred energy source for *Butyrivibrio fibrisolvens* [47]. *Butyrivibrio fibrisolvens* efficiently hydrolyzes these structural polysaccharides and produces butyrate as a primary metabolic end-product [48].

Decreasing the dietary NDF level significantly reduced total CH₄ concentrations, CH₄ concentration per DMI and ADG (ppm/kg) from both respiration and eructation (Table 6; $P < 0.05$). These findings are consistent with previous studies [16,18,19] reporting that increasing NFC and decreasing NDF lead to reduced CH₄ concentrations. In this study, the reduction in NDF level shifted the fermentation pattern towards lower acetate and higher propionate proportions, resulting in a significantly decreased A:P ratio ($P < 0.05$). These shifts may help explain the lower CH₄ concentrations, although hydrogen fluxes and microbial populations

were not measured. Stoichiometrically, the fermentation pathway yielding propionate consumes metabolic H_2 , whereas the pathway producing acetate releases H_2 , the principal substrate for methanogenesis [14]. Consequently, the shift toward propionate production reduces the total H_2 pool available to methanogens, thereby limiting CH_4 synthesis. Furthermore, the reduction in fiber digestibility observed at lower NDF levels supports this mechanism. The reduced fiber digestion may be consistent with lower activity of fibrolytic bacteria, which are the major producers of acetate and the primary suppliers of H_2 to methanogens [49]. However, no significant differences in ruminal pH were detected among treatments in the present study ($P > 0.05$). These results may be attributable to the limitations of the rumen sampling method used in this study. The oral stomach tube technique employed for ruminal sampling may have reduced sensitivity in detecting low pH conditions [50], and pH was measured only from ruminal fluid collected at three time points (−1, +3, and +6 h after the morning feeding). Consequently, this approach could not capture the full 24-hour fluctuations in ruminal pH.

In this study, not only the dietary NDF level but also the NDF source significantly affected CH_4 concentrations (Table 6; $P < 0.05$). Interestingly, despite maintaining an NDF level similar to that of the HFHC group and showing no significant difference ($P > 0.05$) in the A:P ratio—a key indicator of the stoichiometric potential for CH_4 production—the LFLC group exhibited significantly lower CH_4 concentrations from both respiration and eructation than the HFHC group (Table 6; $P < 0.05$). One possible explanation for these results may be related to differences in the physical characteristics of forage- and concentrate-derived fiber. Concentrate-derived fiber may have a smaller particle size and greater functional specific gravity than forage-derived fiber, which could affect ruminal mat formation and passage kinetics [21]. Such physical attributes could facilitate movement of particles out of the ruminal

mat and increase passage through the reticulo-omasal orifice [18]. Faster passage could shorten the time available for fermentation, but similar fiber digestibility in LFLC and HFHC (Table 5; $P > 0.05$) does not establish either faster passage or faster degradation. A possible difference between the time required for fiber degradation and for methanogen colonization and syntrophic H_2 transfer has been proposed in the literature [48,49]; its contribution to the present concentration-based responses remains hypothetical. [21,51,52] Moreover, the significant difference in CH_4 concentrations from respiration and eructation resulted in a significant difference in CH_4 concentration per ADG (ppm/kg), even though the NDF source did not significantly affect ADG (Table 6; $P < 0.05$). Interestingly, however, while DMI, like ADG, was not significantly influenced by the NDF source, the NDF source did not exert a significant effect on CH_4 concentration per DMI (ppm/kg; Table 6; $P > 0.05$). The different statistical outcomes for CH_4 concentration per DMI and per ADG (ppm/kg) may reflect variation in their denominators and associated uncertainty. A nonsignificant difference in the DMI-based ratio does not demonstrate equivalent hydrogen production or methane-generating capacity of the two fiber sources. Furthermore, the moderate reduction in total CH_4 concentrations in the LFLC group was partially offset by a numerical increase in DMI (i.e., a 4% increase relative to HFHC), rendering the reduction in CH_4 concentration per DMI (ppm/kg) statistically indistinguishable given individual animal variations. In contrast, CH_4 concentration per ADG (ppm/kg) exhibited a significant decline. Because CH_4 concentration per ADG (ppm/kg) is expressed as the ratio of CH_4 concentration to ADG, the lower value observed in LFLC occurred concurrently with significantly lower CH_4 concentration from respiration and eructation and a numerically 7% higher ADG than HFHC. However, the difference in ADG was not statistically significant and therefore cannot be considered evidence of improved growth performance. Accordingly, the significant reduction in CH_4 concentration per ADG (ppm/kg) should be interpreted as lower

CH₄ concentration per unit of ADG, rather than as evidence of a biological interaction or synergistic effect between CH₄ concentration and growth performance. The numerical difference in ADG warrants further investigation in studies with greater statistical power. A major limitation is that NDF was analyzed only in the complete TMR, not in the individual ingredients. Thus, actual fNDF and cNDF concentrations and their proportions of total NDF could not be quantified. Particle-size distribution, peNDF, functional specific gravity, ruminal mat formation, and passage rate were not measured. Mechanisms involving these characteristics therefore remain hypotheses. In addition, differences in palm kernel meal, urea, and other ingredients and in energy and nutrient composition prevent the responses from being attributed exclusively to NDF level or source. These accompanying differences, including the non-protein nitrogen supplied by urea in LFLC, may have contributed to the fermentation responses.

CONCLUSIONS

In conclusion, dietary NDF level and source were associated with changes in ruminal fermentation and CH₄ concentration responses without significantly affecting growth performance in growing Hanwoo heifers. Lower dietary NDF was associated with increased total VFA concentration, decreased acetate proportion and acetate-to-propionate ratio, increased propionate and butyrate proportions, and lower fiber digestibility. These responses were accompanied by lower CH₄ concentration per DMI and ADG. In the NDF-source contrast, LFLC exhibited lower acetate and propionate proportions and a higher butyrate proportion than HFHC, while nutrient digestibility did not differ between the two diets. LFLC also exhibited lower CH₄ concentration per ADG than HFHC, although CH₄ concentration per DMI did not differ between the two diets. Overall, lower dietary NDF levels showed a more consistent

association with lower CH₄ concentration-based responses, whereas differences in NDF source were primarily reflected in ruminal fermentation characteristics and CH₄ concentration per ADG. These associations should be interpreted within the dietary contrasts tested because NDF characteristics were not separated from accompanying differences in ingredients and nutrient supply.

CONFLICT OF INTEREST

We certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript. Oh JP is an employee of Cargill Animal Nutrition Korea.

AUTHORS' CONTRIBUTION

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Not applicable.

SUPPLEMENTARY MATERIAL

Not applicable.

DATA AVAILABILITY

Upon reasonable request, the datasets of this study can be available from the corresponding author.

ETHICS APPROVAL

The use of animals and protocols for this experiment were reviewed and pre-approved by the Chungnam National University Animal Research Ethics Committee (202109A-CNU-108).

DECLARATION OF GENERATIVE AI

During the preparation of this work, Gemini 3 was used for the purpose of English language editing. After using this tool, the manuscript was reviewed and edited as needed and authors will assume full responsibility for the publication.

Reference

1. Tseten T, Sanjorjo RA, Kwon M, TT S-WK. Strategies to mitigate enteric methane emissions from ruminant animals. *J. Microbiol. Biotechnol.* 2022;32:269-77. <https://doi.org/10.4014/jmb.2202.02019>
2. Johnson KA, Johnson DE. Methane emissions from cattle. *J. Anim. Sci.* 1995;73:2483-92. <https://doi.org/10.2527/1995.7382483x>
3. Arndt C, Hristov AN, Price WJ, McClelland SC, Pelaez AM, Cueva SF, Oh J, Dijkstra J, Bannink A, Bayat AR. Full adoption of the most effective strategies to mitigate methane emissions by ruminants can help meet the 1.5° C target by 2030 but not 2050. *Proc. Natl. Acad. Sci. USA.* 2022;119:e2111294119. <https://doi.org/10.1073/pnas.2111294119>
4. Lee H, Suetomi Y, Haga S, Roh S. Ruminal methanogen abundance and associated physiological characteristics in Japanese Black–Holstein F1 calves. *Anim. Biosci.* 2026;39:260205. <https://doi.org/10.5713/ab.260205>
5. Haque MN. Dietary manipulation: a sustainable way to mitigate methane emissions from ruminants. *J. Anim. Sci. Technol.* 2018;60:1-10. <https://doi.org/10.1186/s40781-018-0175-7>
6. Hristov AN, Melgar A, Wasson D, Arndt C. Symposium review: Effective nutritional strategies to mitigate enteric methane in dairy cattle. *J. Dairy Sci.* 2022. <https://doi.org/10.3168/jds.2021-21398>
7. Shinkai T, Takizawa S, Fujimori M, Mitsumori M. — Invited Review — The role of rumen microbiota in enteric methane mitigation for sustainable ruminant production. *Anim Biosci* 2024;37:360-9. <https://doi.org/10.5713/ab.23.0301>

- 577 8. Palangi V, Taghizadeh A, Abachi S, Lackner M. Strategies to mitigate enteric methane
578 emissions in ruminants: A Review. Sustainability 2022;14:13229.
579 <https://doi.org/10.3390/su142013229>
- 580 9. Van Lingen HJ, Niu M, Kebreab E, Valadares Filho SC, Rooke JA, Duthie C-A,
581 Schwarm A, Kreuzer M, Hynd PI, Caetano M. Prediction of enteric methane production,
582 yield and intensity of beef cattle using an intercontinental database. Agric. Ecosyst.
583 Environ. 2019;283:106575. <https://doi.org/10.1016/j.agee.2019.106575>
- 584 10. Wang M, Sun X, Janssen P, Tang S, Tan Z. Responses of methane production and
585 fermentation pathways to the increased dissolved hydrogen concentration generated by
586 eight substrates in in vitro ruminal cultures. Anim. Feed Sci. Technol. 2014;194:1-11.
587 <https://doi.org/10.1016/j.anifeedsci.2014.04.012>
- 588 11. Ungerfeld EM. Shifts in metabolic hydrogen sinks in the methanogenesis-inhibited
589 ruminal fermentation: a meta-analysis. Front. Microbiol. 2015;6:37.
590 <https://doi.org/10.3389/fmicb.2015.00037>
- 591 12. Mackie RI, Kim H, Kim NK, Cann I. — Invited Review — Hydrogen production and
592 hydrogen utilization in the rumen: key to mitigating enteric methane production. Anim
593 Biosci 2024;37:323-36. <https://doi.org/10.5713/ab.23.0294>
- 594 13. Beauchemin K, McGinn S. Methane emissions from feedlot cattle fed barley or corn
595 diets. J. Anim. Sci. 2005;83:653-61. <https://doi.org/10.2527/2005.833653x>
- 596 14. Janssen PH. Influence of hydrogen on rumen methane formation and fermentation
597 balances through microbial growth kinetics and fermentation thermodynamics. Anim.
598 Feed Sci. Technol. 2010;160:1-22. <https://doi.org/10.1016/j.anifeedsci.2010.07.002>

- 599 15. Newbold CJ, De la Fuente G, Belanche A, Ramos-Morales E, McEwan NR. The role
600 of ciliate protozoa in the rumen. *Front. Microbiol.* 2015;6:1313.
601 <https://doi.org/10.3389/fmicb.2015.01313>
- 602 16. Aguerre MJ, Wattiaux MA, Powell J, Broderick GA, Arndt C. Effect of forage-to-
603 concentrate ratio in dairy cow diets on emission of methane, carbon dioxide, and
604 ammonia, lactation performance, and manure excretion. *J. Dairy Sci.* 2011;94:3081-93.
605 <https://doi.org/10.3168/jds.2010-4011>
- 606 17. Olijhoek D, Løvendahl P, Lassen J, Hellwing A, Höglund J, Weisbjerg M, Noel S,
607 McLean F, Højberg O, Lund P. Methane production, rumen fermentation, and diet
608 digestibility of Holstein and Jersey dairy cows being divergent in residual feed intake
609 and fed at 2 forage-to-concentrate ratios. *J. Dairy Sci.* 2018;101:9926-40.
610 <https://doi.org/10.3168/jds.2017-14278>
- 611 18. Lovett D, Lovell S, Stack L, Callan J, Finlay M, Conolly J, O'mara F. Effect of
612 forage/concentrate ratio and dietary coconut oil level on methane output and
613 performance of finishing beef heifers. *Livest. Prod. Sci.* 2003;84:135-46.
614 <https://doi.org/10.1016/j.livprodsci.2003.09.010>
- 615 19. Kang K, Cho H, Jeong S, Jeon S, Lee M, Lee S, Baek Y, Oh J, Seo S. Application of a
616 hand-held laser methane detector for measuring enteric methane emissions from cattle
617 in intensive farming. *J. Anim. Sci.* 2022;100:skac211.
618 <https://doi.org/10.1093/jas/skac211>
- 619 20. Bradford B, Mullins C. Invited review: Strategies for promoting productivity and health
620 of dairy cattle by feeding nonforage fiber sources. *J. Dairy Sci.* 2012;95:4735-46.
621 <https://doi.org/10.3168/jds.2012-5393>

21. Grant R. Interactions among forages and nonforage fiber sources. *J. Dairy Sci.* 1997;80:1438-46. [https://doi.org/10.3168/jds.S0022-0302\(97\)76073-9](https://doi.org/10.3168/jds.S0022-0302(97)76073-9)
22. Seo S, Jeon S, Ha JK. Guidelines for experimental design and statistical analyses in animal studies submitted for publication in the Asian-Australasian Journal of Animal Sciences. *Asian-australas. J. Anim. Sci.* 2018;31:1381. <https://doi.org/10.5713/ajas.18.0468>
23. NIAS. Korean Feeding Standard for Hanwoo. 3 rd rev. ed. Jeonju. National Institute of Animal Science in Rural Development Administration; 2017.
24. Lee M, Jeong S, Seo J, Seo S. Changes in the ruminal fermentation and bacterial community structure by a sudden change to a high-concentrate diet in Korean domestic ruminants. *Asian-australas. J. Anim. Sci.* 2019;32:92. <https://doi.org/10.5713/ajas.18.0262>
25. Oh J, Cho H, Kang N, Lee M, Hoque MR, Kim Y, Seo S. Effects of dietary crude protein levels on growth performance, rumen characteristics, blood metabolites, and methane emissions in finishing Hanwoo steers. *Anim. Biosci.* 2025;38. <https://doi.org/10.5713/ab.25.0063>
26. Jeon S, Sohn K-N, Seo S. Evaluation of feed value of a by-product of pickled radish for ruminants: analyses of nutrient composition, storage stability, and in vitro ruminal fermentation. *J. Anim. Sci. Technol.* 2016;58:1-9. <https://doi.org/10.1186/s40781-016-0117-1>
27. NRC. Nutrient requirements of dairy cattle. 7th ed. Washington, DC. National Academies Press; 2001.
28. NRC. Nutrient requirements of beef cattle. 7th ed. Washington, DC. National Academies Press; 2000.

29. Fox D, Tedeschi L, Tylutki T, Russell J, Van Amburgh M, Chase L, Pell A, Overton T. The Cornell Net Carbohydrate and Protein System model for evaluating herd nutrition and nutrient excretion. *Anim. Feed Sci. Technol.* 2004;112:29-78. <https://doi.org/10.1016/j.anifeedsci.2003.10.006>
30. Van Soest Pv, Robertson JB, Lewis BA. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. *J. Dairy Sci.* 1991;74:3583-97. [https://doi.org/10.3168/jds.S0022-0302\(91\)78551-2](https://doi.org/10.3168/jds.S0022-0302(91)78551-2)
31. Van Keulen J, Young B. Evaluation of acid-insoluble ash as a natural marker in ruminant digestibility studies. *J. Anim. Sci.* 1977;44:282-7. <https://doi.org/10.2527/jas1977.442282x>
32. R Core Team. R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing; 2021. 2021.
33. Allen MS. Effects of diet on short-term regulation of feed intake by lactating dairy cattle. *J. Dairy Sci.* 2000;83:1598-624. [https://doi.org/10.3168/jds.S0022-0302\(00\)75030-2](https://doi.org/10.3168/jds.S0022-0302(00)75030-2)
34. Chen H, Wang C, Huasai S, Chen A. Effects of dietary forage to concentrate ratio on nutrient digestibility, ruminal fermentation and rumen bacterial composition in Angus cows. *Sci. Rep.* 2021;11:17023. <https://doi.org/10.1038/s41598-021-96580-5>
35. Conrad H, Pratt A, Hibbs JW. Regulation of feed intake in dairy cows. I. Change in importance of physical and physiological factors with increasing digestibility. *J. Dairy Sci.* 1964;47:54-62. [https://doi.org/10.3168/jds.S0022-0302\(64\)88581-7](https://doi.org/10.3168/jds.S0022-0302(64)88581-7)
36. Serment A, Schmidely P, Giger-Reverdin S, Chapoutot P, Sauvant D. Effects of the percentage of concentrate on rumen fermentation, nutrient digestibility, plasma metabolites, and milk composition in mid-lactation goats. *J. Dairy Sci.* 2011;94:3960-72. <https://doi.org/10.3168/jds.2010-4041>

- 670 37. Mertens D. Creating a system for meeting the fiber requirements of dairy cows. J. Dairy
671 Sci. 1997;80:1463-81. [https://doi.org/10.3168/jds.S0022-0302\(97\)76075-2](https://doi.org/10.3168/jds.S0022-0302(97)76075-2)
- 672 38. Zebeli Q, Aschenbach JR, Tafaj M, Boguhn J, Ametaj B, Drochner W. Invited review:
673 Role of physically effective fiber and estimation of dietary fiber adequacy in high-
674 producing dairy cattle. J. Dairy Sci. 2012;95:1041-56. [https://doi.org/10.3168/jds.2011-](https://doi.org/10.3168/jds.2011-4421)
675 [4421](https://doi.org/10.3168/jds.2011-4421)
- 676 39. Owens F, Goetsch A. Digesta passage and microbial protein synthesis. Pages 196-223
677 in Control of Digestion and Metabolism in Ruminants. L. P. Milligan, W. L. Grovum,
678 and A. Dobson ed. Prentice-Hall, Englewood Cliffs, NJ. 1986.
- 679 40. Lukas M, Sudekum K-H, Rave G, Friedel K, Susenbeth A. Relationship between fecal
680 crude protein concentration and diet organic matter digestibility in cattle. J. Anim. Sci.
681 2005;83:1332-44. <https://doi.org/10.2527/2005.8361332x>
- 682 41. Goad D, Goad C, Nagaraja T. Ruminal microbial and fermentative changes associated
683 with experimentally induced subacute acidosis in steers. J. Anim. Sci. 1998;76:234-41.
684 <https://doi.org/10.2527/1998.761234x>
- 685 42. Nagaraja T, Titgemeyer E. Ruminal acidosis in beef cattle: the current microbiological
686 and nutritional outlook. J. Dairy Sci. 2007;90:E17-E38.
687 <https://doi.org/10.3168/jds.2006-478>
- 688 43. Moorby JM, Dewhurst RJ, Evans RT, Danelon J. Effects of dairy cow diet forage
689 proportion on duodenal nutrient supply and urinary purine derivative excretion. J. Dairy
690 Sci. 2006;89:3552-62. [https://doi.org/10.3168/jds.S0022-0302\(06\)72395-5](https://doi.org/10.3168/jds.S0022-0302(06)72395-5)
- 691 44. Abijaoudé J, Morand-Fehr P, Tessier J, Schmidely P, Sauvant D. Influence of forage:
692 concentrate ratio and type of starch in the diet on feeding behaviour, dietary preferences,

- digestion, metabolism and performance of dairy goats in mid lactation. Anim. Sci. 2000;71:359-68. <https://doi.org/10.1017/S1357729800055211>
45. Murphy M, Baldwin R, Koong L. Estimation of stoichiometric parameters for rumen fermentation of roughage and concentrate diets. J. Anim. Sci. 1982;55:411-21. <https://doi.org/10.2527/jas1982.552411x>
46. Alimon AR. The nutritive value of palm kernel cake for animal feed. Palm. Oil. Dev. 2004;40:12-4.
47. Hespell R, Wolf R, Bothast R. Fermentation of xylans by *Butyrivibrio fibrisolvens* and other ruminal bacteria. Appl. Environ. Microbiol. 1987;53:2849-53. <https://doi.org/10.1128/aem.53.12.2849-2853.1987>
48. Emerson EL, Weimer PJ. Fermentation of model hemicelluloses by *Prevotella* strains and *Butyrivibrio fibrisolvens* in pure culture and in ruminal enrichment cultures. Appl. Microbiol. Biotechnol. 2017;101:4269-78. <https://doi.org/10.1007/s00253-017-8150-7>
49. Russell JB, Wilson DB. Why are ruminal cellulolytic bacteria unable to digest cellulose at low pH? J. Dairy Sci. 1996;79:1503-9. [https://doi.org/10.3168/jds.S0022-0302\(96\)76510-4](https://doi.org/10.3168/jds.S0022-0302(96)76510-4)
50. Duffield T, Plaizier J, Fairfield A, Bagg R, Vessie G, Dick P, Wilson J, Aramini J, McBride B. Comparison of techniques for measurement of rumen pH in lactating dairy cows. J. Dairy Sci. 2004;87:59-66. [https://doi.org/10.3168/jds.S0022-0302\(04\)73142-2](https://doi.org/10.3168/jds.S0022-0302(04)73142-2)
51. McAllister T, Bae H, Jones G, Cheng K-J. Microbial attachment and feed digestion in the rumen. J. Anim. Sci. 1994;72:3004-18. <https://doi.org/10.2527/1994.72113004x>

- 715 52. Goopy JP, Donaldson A, Hegarty R, Vercoe PE, Haynes F, Barnett M, Oddy VH. Low-
716 methane yield sheep have smaller rumens and shorter rumen retention time. Br. J. Nutr.
717 2014;111:578-85. <https://doi.org/10.1017/S0007114513002936>
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Table 1. Diet formulation of the experimental diets

Ingredients (g/kg DM) ¹⁾	Treatments ²⁾		
	HFHC	LFLC	LFHC
Oat hay	50	30	249
Perennial ryegrass	354	274	50
Corn, ground	111	36	111
Wheat, ground	43	57	42
Corn gluten feed	354	191	458
Rice bran	6	18	6
Molasses	11	8	11
Palm kernel meal	38	331	38
Urea	0	9	0
CMS	11	15	11
Ammonium chloride	0	1	0
Limestone	20	26	20
Salt	1	0	1
Vitamin and mineral mix ³⁾	2	2	2

¹⁾CMS, condensed molasses solubles.

²⁾HFHC, a high forage-to-concentrate ratio (42.5:57.5) with a **high** energy concentrate mix, LFLC, a low forage-to-concentrate ratio (30:70) with a low energy concentrate mix, LFHC, a low forage-to-concentrate ratio (30:70) with a high energy concentrate mix.

³⁾33,330,000 IU/kg vitamin A, 40,000,000 IU/kg vitamin D, 20.86 IU/kg vitamin E, 20 mg/kg Cu, 90 mg/kg Mn, 100 mg/kg Zn, 250 mg/kg Fe, 0.4 mg/kg I, and 0.4 mg/kg Se.

728 **Table 2. Chemical composition** (g/kg DM or as stated) of the experimental diets

Items ¹⁾	Treatments ²⁾		
	HFHC	LFLC	LFHC
DM, g/kg as fed	956	931	939
OM	924	931	930
CP	132	136	131
SOLP	34	43	40
NDICP	11	25	13
ADICP	8	11	7
aNDF	569	557	501
ADF	342	315	276
ADL	43	46	38
Starch	136	148	160
Ether extract	27	34	32
Ash	76	69	70
Ca	6	7	6
P	3	3	4
K	9	9	10
Na	1	2	3
Cl	3	6	5
S	2	2	2
Mg	2	2	3
TDN	613	629	654
NEm, MJ/kg DM	5.6	5.9	6.2
NEg, MJ/kg DM	3.2	3.5	3.7
Total carbohydrates	811	777	793
NFC	253	246	304
Carbohydrate fraction, g/kg carbohydrate			
CA	67	72	115
CB1	168	190	202
CB2	144	126	182
CB3	561	541	502
CC	126	143	114
Protein fraction, g/kg CP			
PA+B1	391	361	381
PB2	477	425	498
PB3	38	123	51
PC	94	91	70

¹⁾DM, dry matter; OM, organic matter; CP, crude protein; SOLP, soluble CP; NDICP, neutral detergent insoluble CP; ADICP, acid detergent insoluble CP; aNDF, neutral detergent fiber analyzed using a heat stable amylase and expressed inclusive of residual ash; ADF, acid detergent fiber; ADL, acid detergent lignin; TDN, total digestible nutrients; NEm, net energy for maintenance; NEg, net energy for growth; NFC, non-fiber carbohydrate. CA, carbohydrate A fraction; ethanol soluble carbohydrates; CB1, carbohydrate B1 fraction; starch; CB2, carbohydrate B2 fraction; soluble fiber; CB3, carbohydrate B3 fraction; available insoluble fiber; CC, carbohydrate C fraction; unavailable carbohydrate. PA+B1, protein A and B1 fractions; soluble CP; PB2, protein B2 fraction; intermediate degradable CP; PB3, protein B3 fraction, slowly degradable fiber-bound CP; PC, protein C fraction, unavailable CP.

²⁾HFHC, a high forage-to-concentrate ratio (42.5:57.5) with a **high** energy concentrate mix, LFLC, a low forage-to-concentrate ratio (30:70) with a low energy concentrate mix, LFHC, a low forage-to-concentrate ratio (30:70) with a high energy concentrate mix. **Total aNDF was analyzed in the complete TMR; ingredient-specific NDF was not measured, so actual fNDF and cNDF concentrations and their proportions of total NDF are unavailable.**

743 **Table 3.** Effects of neutral detergent fiber level and its source of experimental diets on growth performance in growing Hanwoo heifers

Items	Treatments			SEM	<i>P</i> -value		
	HFHC	LFLC	LFHC		Mean	Contrast	
						NDF level	NDF source
Initial BW, kg	296	302	297	8.3	0.879	0.814	0.651
Final BW, kg	359	369	368	11.5	0.803	0.809	0.547
ADG, g/day	750	803	844	62.7	0.546	0.371	0.546
DMI, kg/day	7.17	7.46	8.02	0.334	0.207	0.099	0.554
FCR	10.31	9.31	10.07	0.792	0.673	0.792	0.396

744 BW, body weight; ADG, average daily gain; DMI, dry matter intake; FCR, feed conversion ratio, **DMI (g) / ADG (g)**.
 745 HFHC, a high forage-to-concentrate ratio (42.5:57.5) with a high energy concentrate mix; LFLC, a low forage-to-concentrate ratio (30:70) with
 746 a low energy concentrate mix; LFHC, a low forage-to-concentrate ratio (30:70) with a high energy concentrate mix.
 747 NDF, neutral detergent fiber, NDF level, LFHC vs. the average of HFHC and LFLC; NDF source, HFHC vs. LFLC.

748 **Table 4.** Effects of **neutral detergent fiber level and source** of experimental diets on rumen characteristics in growing Hanwoo heifers

Items	Treatments			SEM	P-value		
	HFHC	LFLC	LFHC		Mean	Contrast	
						NDF level	NDF source
pH	6.4	6.4	6.4	0.07	0.913	0.741	0.781
NH ₃ -N, mg/dL	7.6 ^b	11.1 ^a	11.7 ^a	1.28	0.047	0.134	0.052
Total VFA, mM	67.3	65.6	73.2	2.44	0.062	0.021	0.617
Molar proportions, mmol/mol							
Acetate	669.3 ^a	649.7 ^b	616.8 ^c	5.82	< 0.001	< 0.001	0.018
Propionate	203.1 ^a	187.4 ^b	213.9 ^a	4.09	< 0.001	< 0.001	0.007
Isobutyrate	12.9 ^b	15.8 ^a	14.4 ^{ab}	0.59	0.004	0.984	0.001
Butyrate	90.7 ^b	116.6 ^a	122.1 ^a	2.83	< 0.001	< 0.001	< 0.001
Isovalerate	13.0 ^b	17.8 ^a	16.2 ^a	0.86	< 0.001	0.430	< 0.001
Valerate	11.0 ^b	12.9 ^b	16.6 ^a	0.71	< 0.001	< 0.001	0.065
Acetate:Propionate	3.3 ^a	3.5 ^a	2.9 ^b	0.09	< 0.001	< 0.001	0.128

749 **NH₃-N, ammonia; VFA, Volatile fatty acid.**

750 HFHC, a high forage-to-concentrate ratio (42.5:57.5) with a high energy concentrate mix; LFLC, a low forage-to-concentrate ratio (30:70) with
 751 a low energy concentrate mix; LFHC, a low forage-to-concentrate ratio (30:70) with a high energy concentrate mix.

752 NDF, neutral detergent fiber, NDF level, LFHC vs. the average of HFHC and LFLC; NDF source, HFHC vs. LFLC.

753 ^{a-b}Means within the same row with different superscripts are significantly different ($P < 0.05$).

754 **Table 5.** Effects of **neutral detergent fiber level and source** of experimental diets on digestibility in growing Hanwoo heifers

Items	Treatments			SEM	P-value		
	HFHC	LFLC	LFHC		Mean	Contrast	
						NDF level	NDF source
DM, %	66.06	62.42	64.36	1.303	0.185	0.941	0.072
OM, %	68.51	65.57	68.07	1.240	0.235	0.512	0.119
CP, %	67.44 ^a	66.83 ^{ab}	62.14 ^b	1.390	0.038	0.013	0.760
EE, %	74.00	80.15	70.61	2.598	0.065	0.065	0.120
aNDF, %	64.14 ^a	61.79 ^{ab}	55.61 ^b	2.019	0.030	0.012	0.427
ADF, %	59.65 ^a	53.83 ^{ab}	50.77 ^b	2.272	0.048	0.053	0.095

755 DM, dry matter; OM, organic matter; CP, crude protein; EE, ether extract; aNDF, neutral detergent fiber analyzed using a heat stable amylase
 756 and expressed inclusive of residual ash; ADF, acid detergent fiber.

757 HFHC, a high forage-to-concentrate ratio (42.5:57.5) with a high energy concentrate mix; LFLC, a low forage-to-concentrate ratio (30:70) with
 758 a low energy concentrate mix; LFHC, a low forage-to-concentrate ratio (30:70) with a high energy concentrate mix.

759 NDF level, LFHC vs. the average of HFHC and LFLC; NDF source, HFHC vs. LFLC.

760 ^{a-b}Means within the same row with different superscripts are significantly different ($P < 0.05$).

761 **Table 6.** Effects of **neutral detergent fiber level and source** of experimental diets on methane concentration in growing Hanwoo heifers

Items	Treatments			SEM	P-value		
	HFHC	LFLC	LFHC		Mean	Contrast	
						NDF level	NDF source
CH ₄ from respiration							
ppm	24.3 ^a	19.7 ^b	16.4 ^b	1.09	< 0.001	< 0.001	0.005
ppm / kg of DMI	3.1 ^a	2.7 ^{ab}	2.1 ^b	0.25	0.016	0.008	0.266
ppm / kg of ADG	34.9 ^a	25.0 ^b	21.1 ^b	2.79	0.003	0.012	0.016
CH ₄ from eructation							
ppm	121.7 ^a	98.5 ^b	82.7 ^b	6.72	0.001	0.002	0.019
ppm / kg of DMI	15.8 ^a	13.7 ^{ab}	10.5 ^b	1.45	0.036	0.019	0.289
ppm / kg of ADG	178.9 ^a	125.1 ^{ab}	107.7 ^b	17.68	0.016	0.042	0.036

762 CH₄, methane; DMI, dry matter intake; ADG, average daily gain.

763 HFHC, a high forage-to-concentrate ratio (42.5:57.5) with a high energy concentrate mix; LFLC, a low forage-to-concentrate ratio (30:70) with
 764 a low energy concentrate mix; LFHC, a low forage-to-concentrate ratio (30:70) with a high energy concentrate mix. **The LMD output is path-**
 765 **integrated concentration (ppm·m); at a 1-m path length, the numerically equivalent path-averaged concentration is expressed in ppm. Ratios per**
 766 **DMI and ADG are concentration-based indices, not mass-based methane yield or intensity.**

767 NDF, neutral detergent fiber, NDF level, LFHC vs. the average of HFHC and LFLC; NDF source, HFHC vs. LFLC.

768 ^{a-b}Means within the same row with different superscripts are significantly different ($P < 0.05$).